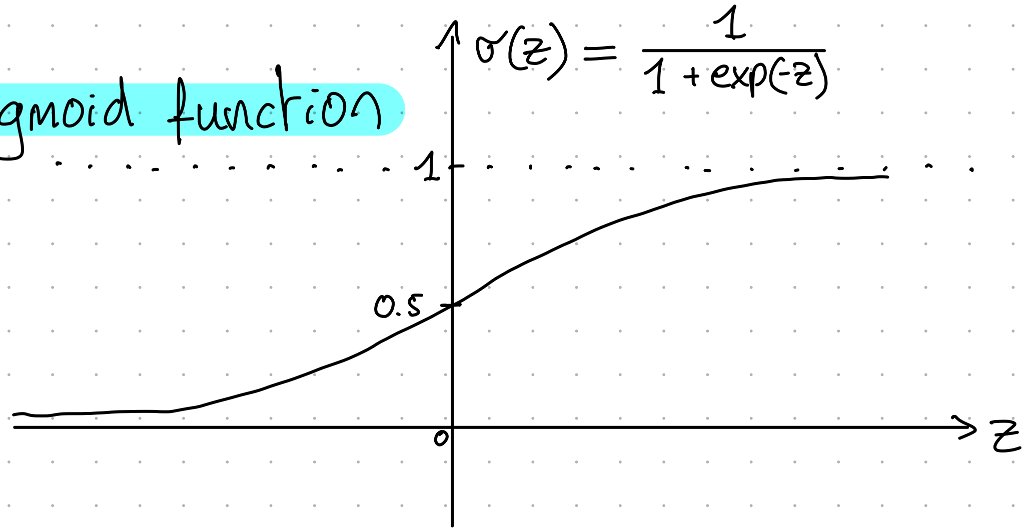
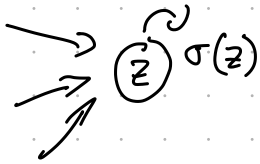
ML Exercises 11

Sigmoid function

$$\sigma(z) = \frac{1}{1+\exp(-z)}$$

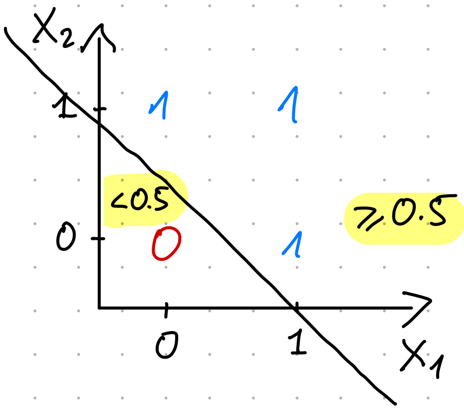


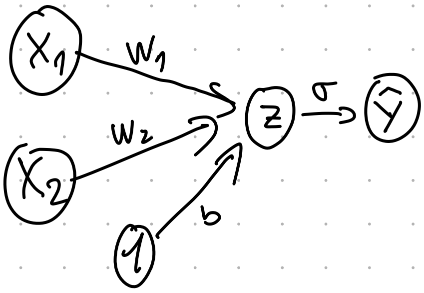


① 0 hidden layers

$\hat{y} = \sigma(z)$

X1	X2	Y	z	$\sigma(z) = \hat{y}$	want
0	0	0	-0.5	<0.5	<0.5
0	1	1	0.5	≥ 0.5	≥ 0.5
1	0	1	0.5	≥ 0.5	≥ 0.5
1	1	1	1.5	≥ 0.5	≥ 0.5





one possible sol:

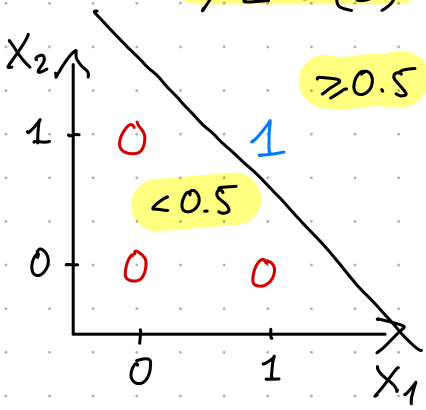
$w_1 = 1 \quad w_2 = 1 \quad b = -0.5$

$z = w_1 x_1 + w_2 x_2 + b$

② 0 hidden layers

$\hat{y} = \sigma(z)$

X1	X2	Y	z	$\sigma(z) = \hat{y}$
0	0	0	$0+b=-1.5$	<0.5
0	1	0	$1+b=-0.5$	<0.5
1	0	0	$1+b=-0.5$	<0.5
1	1	1	$2+b=0.5$	≥ 0.5



$w_1 = 1 \quad w_2 = 1 \quad b = -1.5$

ML Exercises 11

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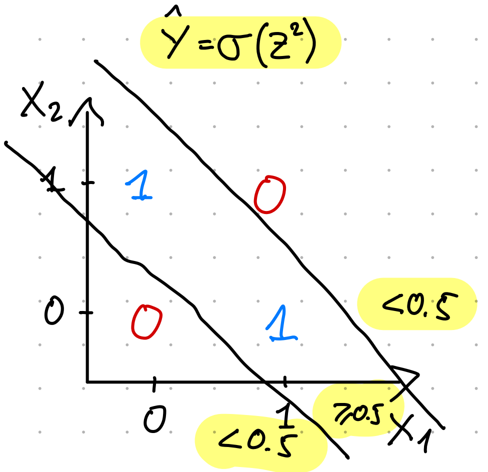
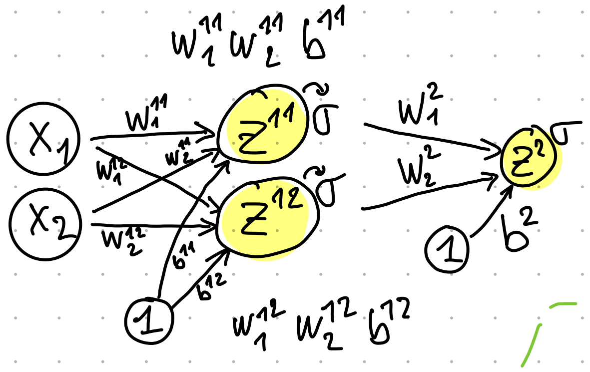
1 hidden layer

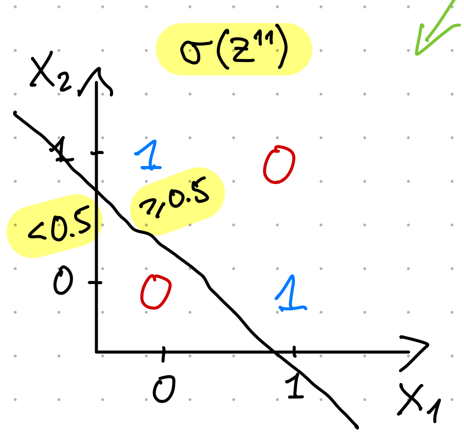
X1	X2	Y	z^{11}	z^{12}	$\sigma(z^{11})$	$\sigma(z^{12})$	z^2	$\hat{y}$
0	0	0	-0.5	1.5	0.38	0.82	$1.20+b^2$	0.495
0	1	1	0.5	0.5	0.62	0.62	$1.24+b^2$	0.505
1	0	1	0.5	0.5	0.62	0.62	$1.24+b^2$	0.505
1	1	0	1.5	-0.5	0.82	0.38	$1.20+b^2$	0.495

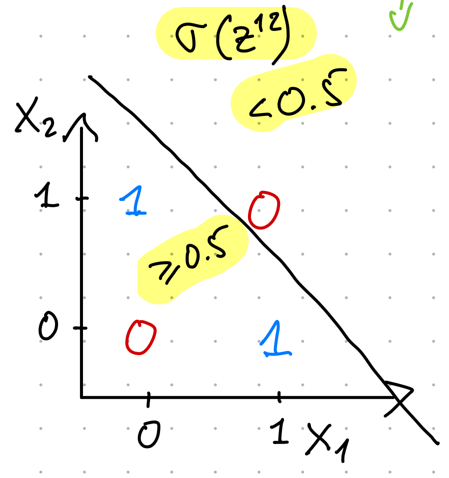




$w_1^2=1 \quad w_2^2=1$

$b^2=-1.22$





$w_1^{11}=1 \quad w_2^{11}=1 \quad b^{11}=-0.5$

$w_1^{12}=-1 \quad w_2^{12}=-1 \quad b^{12}=1.5$